D02 · Demand Planning & Forecasting

02 · Demand sensing & real-time signal integration

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1 WHAT IT IS

Demand sensing replaces the lagging order/shipment signal with near-real-time downstream data — POS, channel inventory, e-commerce, web traffic, weather — feeding a short-horizon model that detects demand turning points in days, not forecast cycles, without overwriting the statistical mid- and long-term baseline.

2 WHY IT MATTERS

Lee, Padmanabhan & Whang 1997 **(Lee et al., 1997 — peer-reviewed)** identified demand-signal processing as a primary cause of the bullwhip effect: when the planning signal is the order stream, each echelon amplifies variance, routinely **2–4× per tier**. The order stream is not demand — it is the customer's reaction to their own forecast and ordering policy, lagged and distorted. Sensing on true downstream data collapses the detection lag from one planning cycle (4–6 weeks) to days, attacking exactly the short-horizon error that drives expedite premiums and reactive excess.

3 HOW IT IS DONE

1. **Map the latency of every signal you have.** POS lands in 1–2 days, channel inventory weekly, shipments arrive cycle-lagged and laundered. Rank inputs by how early they reveal a turn, not by how easy they are to access.
2. **Build a short-horizon sensing layer, separate from the baseline.** Sensing corrects the next 1–8 weeks; the statistical model still owns the mid and long term. Two models, two jobs — never let sensing rewrite the strategic horizon.
3. **Constrain the ML to level and slope.** High-frequency signals are noisy; let the sensing model adjust the near-term level and trend, not invent structure. A sensing model with too much freedom chases noise and amplifies the very variance you came to dampen.
4. **Reconcile the sensed near-term with the statistical mid-term.** Produce one blended forecast, weighted by horizon — sensing-heavy at week 1, baseline-heavy by week 12 — so replenishment sees a single coherent number.
5. **Prove the lift at lag-1 vs lag-4.** Measure forecast error one week out against the naive cycle forecast. If sensing does not beat the cycle forecast at short horizon, the signals are not adding value and you are buying telemetry, not accuracy.
6. **Close the loop to replenishment.** A sensed signal that never changes an order is decoration. Wire the short-horizon correction into the replenishment trigger so detection becomes action automatically.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A CPG supplier planned against retailer orders — **baseline: 1-week-ahead MAPE 28%, $310K/week expedite spend, 3.1 weeks finished-goods cover**. It piped retailer POS and channel inventory into an 8-week sensing layer, blended by horizon, wired to replenishment. Results: **1-week MAPE 28% -> 16% (-43%), expedite spend $310K -> $190K/week (-39%), cover 3.1 -> 2.6 weeks** — with zero change to the 12-month statistical baseline, because sensing fixed the near term and left the strategy alone.

IN PRACTICE

Horizon-weighted blending

Blend the sensed and statistical forecasts with weights that decay by horizon: sensing dominates week 1, the baseline dominates by the planning horizon. The crossover point is an empirical decision — fit it to where sensing stops beating the baseline.

Lag-1 vs lag-4 error measurement

Report short-horizon error separately from total error. Sensing earns its cost only if it improves lag-1 and lag-2 accuracy; the mid-horizon numbers belong to the baseline and should not move. Splitting the metric keeps the two models honest.

Signal governance

Every signal must earn its place with measured predictive lift. Weather, web traffic, and macro indices are seductive and often useless. Add signals one at a time, measure the incremental accuracy, and drop the ones that do not pay.

Closing the loop to replenishment

Connect the sensed correction directly to replenishment and short-term supply. The value of detecting a turn early is exactly zero until it changes an order quantity or a build plan. Automate the handoff; manual reconciliation re-introduces the lag you eliminated.

WHAT YOU WOULD USE

DEMAND SENSING PLATFORMS

SAP IBP Demand Sensing / o9 / Kinaxis — enterprise tier-1 — Native short-horizon sensing layered on the planning baseline, with downstream signal ingestion and automatic blending. **(SAP, o9, Kinaxis — vendor)** The choice when sensing must feed an integrated end-to-end plan.

Blue Yonder Luminate Demand Edge — ML-driven demand sensing with strong retail/CPG signal libraries. **(Blue Yonder — vendor)** Particularly well-suited where POS and channel data are rich and the near-term replenishment loop is the prize.

Specialized sensing / control-tower point solutions — Niche ML sensing engines and control towers that bolt onto an existing planning stack. **(specialized vendors — vendor)** The vertical option when the core APS lacks a credible sensing module.

For operations that don't need a platform — A weekly POS or sell-through pull (EDI 852 or a retailer portal export) feeding a simple short-horizon correction in the existing planning tool. The discipline of looking at the downstream signal weekly captures most of the lag reduction before any platform is bought.

4 SO WHAT

Operations that sense on downstream signals see the turn while it is still cheap to act on; operations that plan on orders see it after the expedite is the only option left. The order stream is the slowest, most distorted view of demand a planner can choose — and most planners choose it because it is the easiest to get. The lag between a turn and your detection of it is the budget for every expedite premium you pay.

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